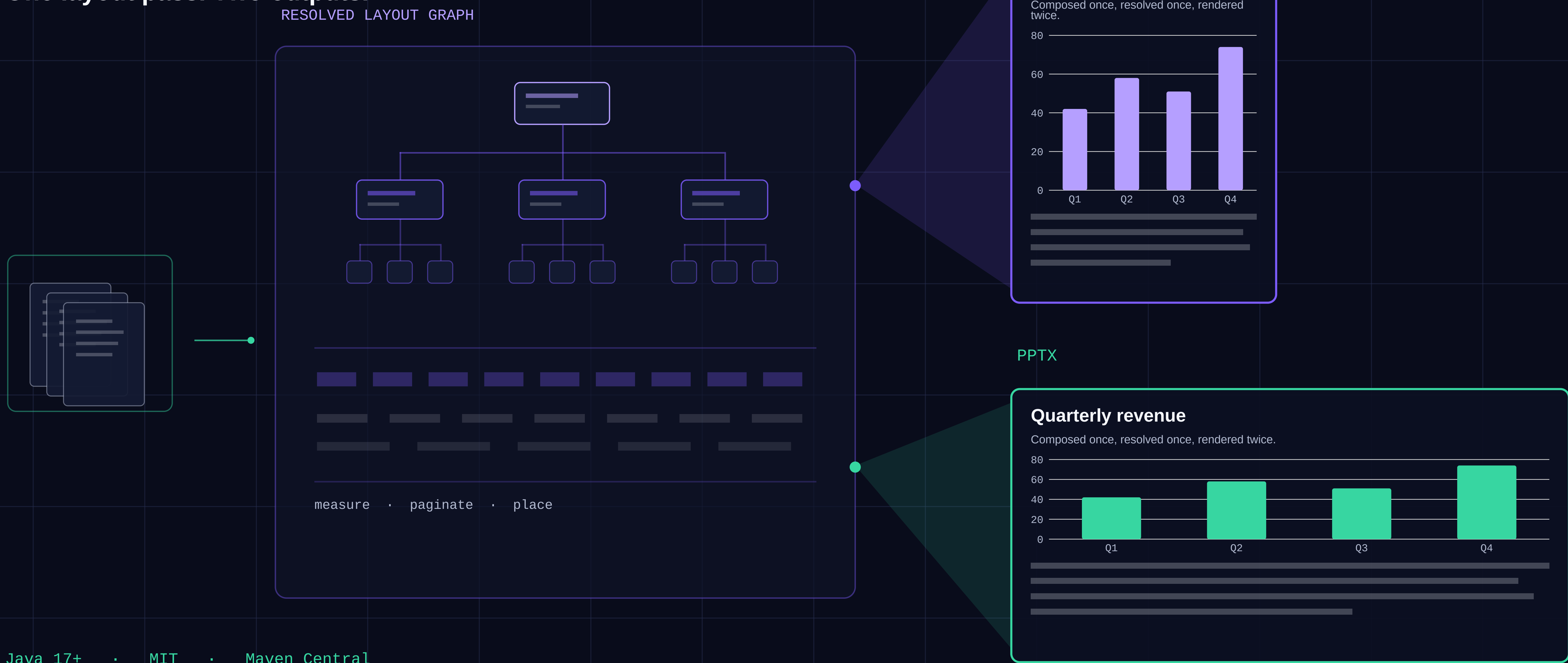
GraphCompose

One layout pass. Two outputs.



Java 17+ · MIT · Maven Central

GRAPHCOMPOSE 2.1

One composition. Two formats.

A Java DSL describes the document. The engine resolves layout and pagination once, then hands the same fragments to every backend — so PDF and PowerPoint are the same geometry, not two hand-kept designs.

PDF	PPTX	DOCX
fixed layout	fixed layout	semantic
PDFBox	Apache POI	Apache POI

This carousel is a GraphCompose document. So is the deck it links to.

WHAT 2.1 ADDS

PowerPoint is a first-class backend.

Fixed layout, not export

PptxFixedLayoutBackend implements the same FixedLayoutRenderer contract as the PDF backend and is discovered through the same ServiceLoader SPI. It consumes placed fragments, so it never re-flows your document.

Native shapes, not screenshots

Text stays text, rectangles stay rectangles, charts stay drawn geometry. The deck opens editable in PowerPoint instead of arriving as a picture of one.

One source, two files

The same compose(...) call writes the PDF and the PPTX. Nothing is authored twice, so the two cannot drift apart.

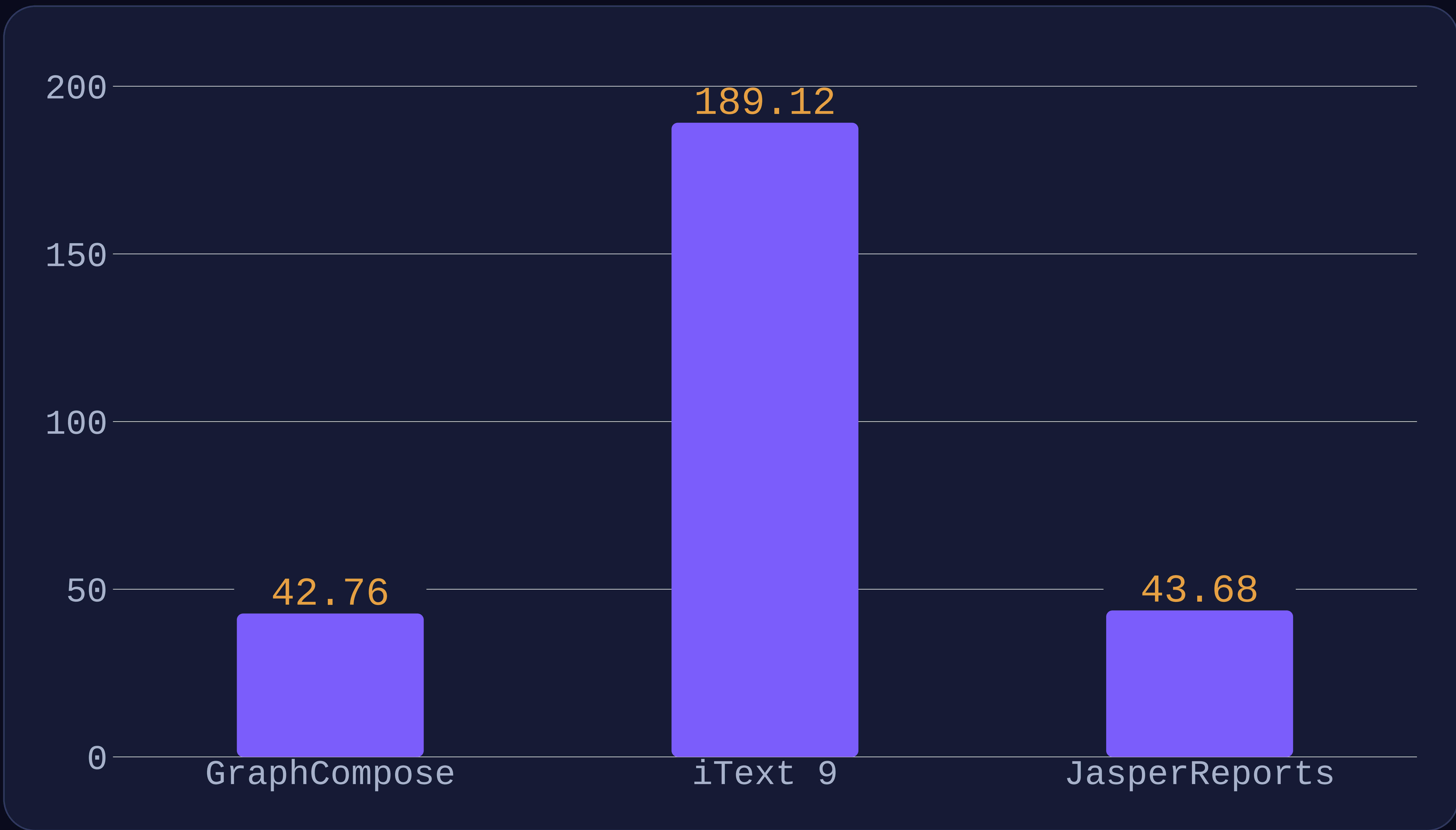
It tells you what it swapped

PowerPoint has no Helvetica or Courier. When a PDF built-in is substituted the render logs the family, the replacement and how to get identical glyphs - rather than silently shifting your text.

MEASURED, NOT CLAIMED

A 1000-row report

Average render latency, milliseconds — lower is faster.



4.4x
faster than iText 9

3.5x
lighter than iText 9

2.7x
lighter than Jasper

One machine, single run, 2026-08-04. Read the ratios rather than the timings: both libraries are measured in the same run, so their proportions survive a change of machine while the milliseconds do not. On time, Jasper reaches parity at this size.

HOW IT STAYS HONEST

Guarantees you can run.

Deterministic layout snapshots

Every flagship document has its geometry recorded as JSON. A layout change that moves a fragment fails the build instead of quietly shipping.

Measurement counts, not stopwatches

A CI gate asserts the exact number of text measurements the layout pass costs. Those are integers, identical on every machine, so they separate a slower engine from a slower laptop — which a millisecond ceiling cannot.

A capability matrix, written down

Where a backend cannot do something, the difference is documented per payload rather than discovered in your output.

Binary compatibility, gated

japicmp checks every build against the published 2.0 baseline, so an accidental break in a minor release fails CI instead of your app.

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Try it.

io.github.demchaav
graph-compose-bundle
latest release on Maven Central

```
try (DocumentSession doc =
        GraphCompose.document(out).create()) {
    doc.pageFlow()
       .addParagraph(p -> p.text("Hello."))
       .build();
    doc.buildPdf();      // or buildPptx(out)
}
```

github.com/DemchaAV/GraphCompose

Java 17+. Every slide in this carousel was composed with the library and rendered by its own PDF backend - the PPTX beside it comes from the same composition.